

Smart Water & Air Quality Monitoring System Using IoT

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1. Problem Understanding

1.1 Background

India faces a severe environmental crisis with both water and air pollution reaching dangerous levels. According to the Central Pollution Control Board (CPCB), over 351 river stretches in India are polluted, and major cities regularly exceed WHO safe limits for air pollutants by 2-3 times.

1.2 Current Challenges

Challenge	Description
Manual Testing	Current methods require lab testing, taking 24-72 hours for results
High Cost	Professional water/air testing costs ₹2,000-5,000 per sample
Delayed Response	By the time results arrive, contamination has already spread
Limited Coverage	Only major cities have monitoring stations; rural areas are ignored
No Real-time Alerts	Communities are unaware of pollution spikes until health issues arise

1.3 Impact

- Health:** Waterborne diseases affect 37.7 million Indians annually
- Economy:** Pollution costs India 2.9% of GDP (~\$560 billion)
- Environment:** Ecosystem destruction, biodiversity loss
- Social:** Vulnerable communities bear disproportionate burden

1.4 Target Users

- Municipal corporations and local bodies
 - Industrial facilities (compliance monitoring)
 - Residential communities and housing societies
 - Agricultural sector (irrigation water quality)
 - Environmental research organizations
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2. Proposed Solution

2.1 Overview

We propose an **IoT-based Smart Environmental Monitoring System** that provides real-time, continuous monitoring of both water and air quality parameters. The system deploys low-cost sensor nodes at multiple locations, transmits data to a cloud platform, and presents insights through an interactive dashboard with automated alerts.

2.2 Key Components

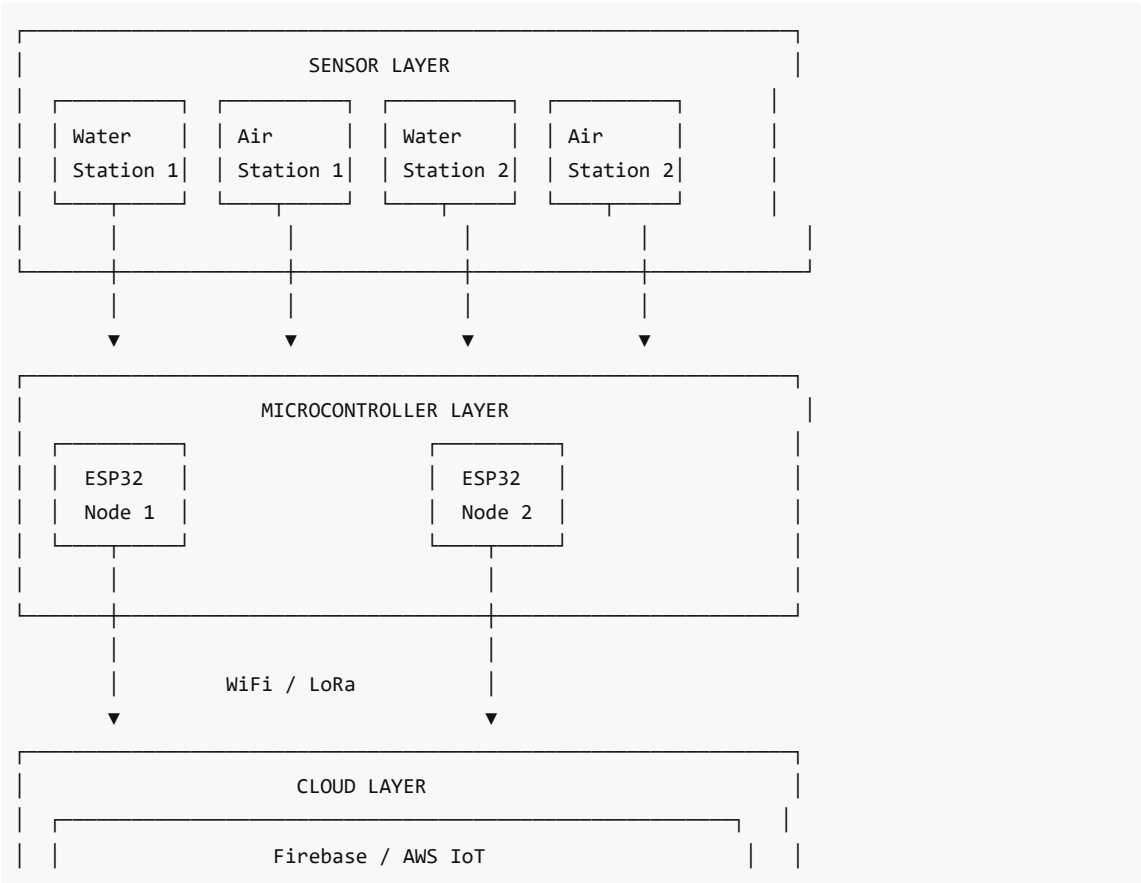
- 1. **Sensor Nodes:** Deploy ESP32 microcontrollers with environmental sensors at monitoring locations
- 2. **Data Transmission:** Use WiFi/LoRa for wireless communication to cloud
- 3. **Cloud Backend:** Store, process, and analyze incoming sensor data
- 4. **Web Dashboard:** Real-time visualization with historical trends
- 5. **Alert System:** Automated notifications when parameters exceed safe thresholds
- 6. **Prediction Engine:** ML-based forecasting of pollution trends

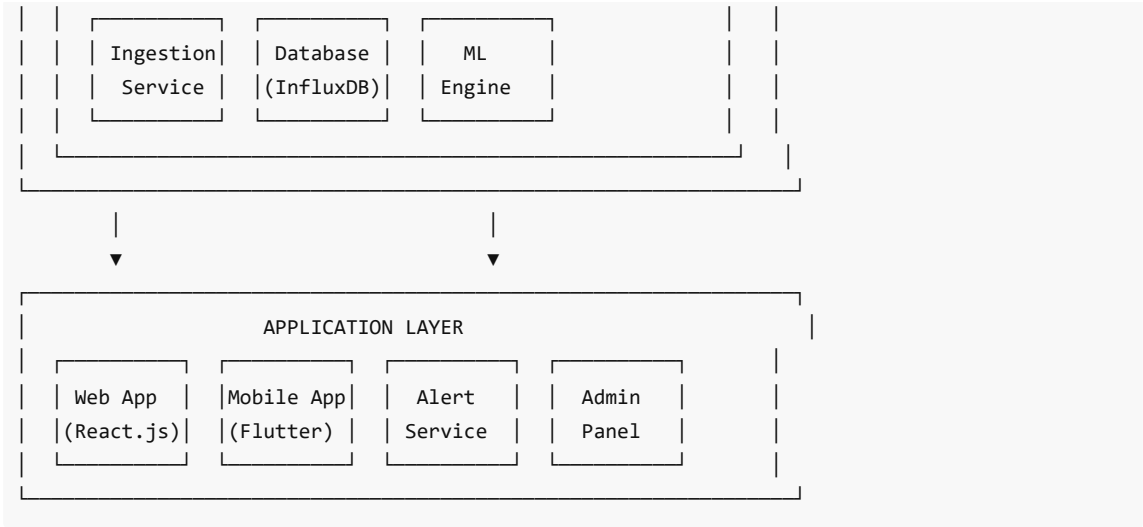
2.3 How It Works

- 1. Sensors continuously measure water quality (pH, turbidity, TDS, dissolved oxygen) and air quality (PM2.5, PM10, CO2, NO2, SO2)
- 2. ESP32 collects data and sends it to cloud via WiFi/LoRa
- 3. Cloud backend processes and stores data in time-series database
- 4. Dashboard displays real-time readings with color-coded status
- 5. When thresholds are breached, alerts are sent via SMS/Email/Push notifications
- 6. Historical data is analyzed to predict future pollution trends

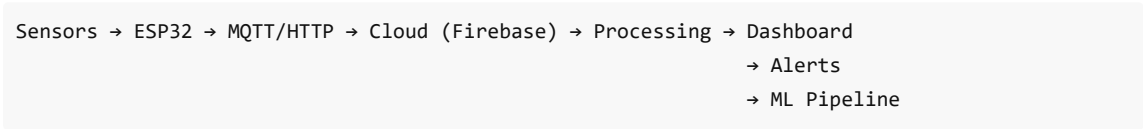
3. System Architecture

3.1 High-Level Architecture





3.2 Data Flow



3.3 Deployment Architecture

- **Edge Devices:** Weatherproof enclosures with solar power option
- **Network:** LoRa gateway for remote areas, WiFi for urban
- **Cloud:** Serverless architecture for cost efficiency
- **CDN:** Static assets served via CDN for fast loading

4. Key Features

4.1 Core Features (MVP)

Feature	Description	Priority
Real-time Monitoring	Live sensor data updates every 30 seconds	P0
Multi-parameter Support	Water: pH, Turbidity, TDS, DO; Air: PM2.5, PM10, CO2, NO2	P0
Interactive Dashboard	Color-coded gauges, charts, and maps	P0
Threshold Alerts	SMS/Email when parameters exceed safe limits	P0
Historical Data	Graphs showing trends over hours/days/weeks	P0
Mobile Responsive	Access dashboard on any device	P0

4.2 Enhanced Features

Feature	Description	Priority
Multi-location	Monitor multiple stations from single dashboard	P1

User Authentication	Role-based access (Admin, Viewer)	P1
Data Export	Download reports as CSV/PDF	P1
Comparative Analysis	Compare data across locations/time periods	P1

4.3 Advanced Features (Future)

Feature	Description	Priority
AI Prediction	Forecast pollution levels for next 24-48 hours	P2
Mobile App	Native Flutter app with push notifications	P2
API Access	REST API for third-party integrations	P2
Smart Recommendations	Actionable insights based on data patterns	P2

5. Technology Stack

5.1 Hardware

Component	Specification	Quantity	Est. Cost
ESP32 DevKit	WiFi + BLE, Dual Core	2	₹800
DHT11 Sensor	Temperature + Humidity	2	₹200
MQ135 Sensor	Air Quality (CO2, NH3, NOx)	2	₹400
pH Sensor Kit	Analog pH 0-14	1	₹1,500
Turbidity Sensor	NTU measurement	1	₹800
TDS Sensor	Total Dissolved Solids	1	₹600
PM2.5 Sensor	Laser dust sensor	1	₹1,200
LoRa Module	SX1276 (optional)	2	₹1,000
Total			₹6,500

5.2 Software Stack

Layer	Technology	Justification
Microcontroller	ESP32 (Arduino IDE / PlatformIO)	WiFi built-in, low power, community support
Communication	MQTT / HTTP	Lightweight, real-time capable
Backend	Node.js + Express	Fast API development, JavaScript ecosystem
Database	Firebase Realtime DB + InfluxDB	Firebase for real-time sync, InfluxDB for time-series
Frontend	React.js + Tailwind CSS	Modern, responsive, fast development

Mobile	Flutter	Cross-platform, single codebase
Cloud	Firebase (Free Tier)	Real-time database, auth, hosting
ML	Python + scikit-learn	Prediction models for pollution trends
Visualization	Chart.js / D3.js	Interactive graphs and gauges

5.3 Development Tools

Tool	Purpose
Git + GitHub	Version control
VS Code	Code editor
Figma	UI/UX design
Postman	API testing
Arduino IDE	Hardware programming

6. Implementation Plan

6.1 Phase Timeline (3 Days)

Day	Morning (9AM-1PM)	Afternoon (2PM-6PM)	Evening (7PM-11PM)
Day 1	Procure components, setup workspace	ESP32 basic setup, sensor connections	Firebase project setup, basic API
Day 2	Sensor calibration, data collection	Cloud integration, data storage	Dashboard UI development
Day 3	Alert system implementation	Testing & bug fixes	Documentation & submission

6.2 Task Distribution

Anirudh (Hardware Lead)

- ☐ Day 1: Component procurement, ESP32 setup
- ☐ Day 1: Sensor wiring and connections
- ☐ Day 2: Sensor calibration and testing
- ☐ Day 2: Data transmission to cloud
- ☐ Day 3: Hardware troubleshooting

Anand (Software Lead)

- ☐ Day 1: Firebase project configuration
- ☐ Day 1: Backend API development
- ☐ Day 2: Dashboard UI with React
- ☐ Day 2: Real-time data visualization

- ☐ Day 3: Alert system, documentation

6.3 Deliverables

Deliverable	Description	Owner
Working Prototype	Functional sensor node + dashboard	Both
GitHub Repository	Clean code with README	Anand
Demo Video	3-minute walkthrough	Both
Documentation	Setup guide, API docs	Anand

7. Innovation & Uniqueness

7.1 What Makes Our Solution Different

1. **Dual Monitoring:** First system to monitor BOTH water and air quality in one integrated platform
2. **Low Cost:** Under ₹7,000 per station vs ₹50,000+ for commercial solutions
3. **Open Source:** All code and designs will be publicly available
4. **Scalable:** LoRa support enables rural deployment without WiFi
5. **Predictive:** ML-based forecasting enables proactive action

7.2 Future Scope

- Integration with government CPCB monitoring network
 - Drone-based sensor deployment for remote areas
 - Blockchain-based data integrity for regulatory compliance
 - Citizen science app for crowdsourced monitoring
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